MILITARY SPECIFICATION

TRUCK, HAND, PLATFORM,

3,000 POUND CAPACITY

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers collapsible, frame, hand, platform trucks.

1.2 Classification. The truck shall be one of the following classes, as specified (see 6.2):

Class 1 - Steel.
Class 2 - Aluminum.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-E-489	- Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

FSC 3920

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-B-26195	- Boxes, Wood-Cleated, Skidded, Load-Bearing Base.

STANDARDS

FEDERAL

FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Double Race or Bi-Level Swivel and Rigid Industrial Casters (ANSI MH 11.1).

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook - SAE J534.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36 - Structural Steel, Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Description. Platform hand trucks (hereinafter called trucks) shall generally consist of a metal frame and deck, hitch, end racks, swivel and rigid casters equipped with pressed or molded on solid rubber tires.

3.2 First article.

3.2.1 Preproduction model. The contractor shall furnish a truck for examination and testing within the time frame specified (see 6.2), to prove that his production methods and choice of design detail will produce trucks that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4, and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein (see 6.2). Material not specified (see 6.4), shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Aluminum. Aluminum material used for construction of the truck shall be selected by the contractor and shall meet all requirements of this specification.

3.3.2 Steel.

3.3.2.1 Angles, plates, bars, and rods. All structural steel angles, plates, bars, and rods shall conform to ASTM A 36.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

3.4 Capacity. The trucks shall have a rated capacity of 3000 pounds plus an overload safety factor of 200 percent.

3.5 Design and construction. Trucks shall be designed and constructed as specified herein. The trucks shall be as specified in Table I. Height dimensions shall be measured from the floor.

Table I. Physical Characteristics of Trucks

	Overall Truck Dimensions (Inches)				
	At Platform (+ 1/2 Inch)			End Rack Height	
Class	Length	Width	Height	Min.	Max.
1 or 2	60	30	10	71	79

3.5.1 Class 1 truck. Class 1 truck shall have the frame, deck, end racks, sockets and caster supports constructed of steel.

3.5.2 Class 2 truck. Class 2 truck shall have the frame, deck, end racks, sockets and caster supports constructed of aluminum.

3.5.3 Frame and deck. The frame and deck may be fabricated from one metal sheet; cut, formed and welded at the corners to form a one-piece construction deck and channel frame; or, the frame and deck may consist of a one-piece metal sheet welded atop a welded frame assembly. The frame assembly shall consist of an outer frame fabricated from formed or structural shapes and similar shaped reinforcing member, if required. The frame and deck shall be a level rigid unit. The frame corners shall be rounded.

3.5.4 Sockets. Sockets for end racks shall be fabricated from metal specified for required truck class. Sockets shall be permanently affixed, one at each corner of the deck. Sixteen additional, equally spaced sockets, shall be provided. Eight sockets shall be located behind each cornerpost of the coupler end of the truck. The sockets shall be designed to provide a stop and support for the end racks. The socket hole shall be between 0.020 and 0.080 inch larger than the outside diameter of the end racks to provide for a snug fit.

3.5.5 Caster support. Caster supports shall consist of specified metal shapes and bars. The supports shall be rigidly attached to the underside of the deck.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

3.5.6 End racks. Two end racks shall be provided for each truck. End racks shall be fabricated from specified metal tubing and shall be reinforced by at least three equally spaced metal crossmembers. The outer frame of the racks shall be U-shaped. The end racks shall be mounted on the sockets of the truck such that they can be removed for easy storage.

3.5.7 Hitch and hitch retainer. A hitch shall be provided for each truck. A hitch retainer shall be provided for each truck, such that the hitch may be retracted and stored within the frame of the truck.

3.5.8 Coupler. A coupler shall be rigidly joined to the rear face of the frame of the truck. The coupler shall be compatible for hookup with the hitch for purposes of towing.

3.5.9 Running gear. Running gear shall consist of two swivel casters and two rigid casters. Casters shall be bolted to the caster supports. Swivel casters and rigid casters shall be medium heavy duty or heavy duty type with dimensions and tolerances in accordance with ANSI MH 11.1. The running gear shall be assembled to the underside of the deck such that an equally distributed load on the deck equal to the rated capacity of the truck will load each caster equally within plus or minus 25 pounds.

3.5.10 Bearings. Bearing load capacity shall withstand the maximum design loadings and tests of Section 4 without exceeding the manufacturer's recommended rating.

3.5.11 Fastening devices. Fastening devices such as bolts, nuts, capscrews, washers, locknuts, cotter pin, and similar parts shall be fabricated from steel. Threads for threaded fasteners shall be in accordance with NBS Handbook H28.

3.5.12 Storage. Hand platform truck shall be designed, such that with the end racks removed, four trucks and their end racks can be stored on a completely assembled truck.

3.6 Performance.

3.6.1 Maximum static loads. The completely assembled truck including casters, wheels, and axles, shall be capable of withstanding without damage and permanent deformation, a static load equal to 200 percent of the rated capacity of the truck uniformly distributed over the entire deck surface. This requirement shall be determined by the test specified in 4.5.2.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

3.6.2 Rolling resistance. Truck, with rated load, shall be capable of being towed with a force not exceeding 160 pounds. This requirement shall be determined by the test specified in 4.5.2.2. The gross weight of the truck shall include the rated load and truck under test.

3.6.3 Dynamic load. Truck shall be capable of being pushed off a 3-inch step to a 1-inch step to the floor at a minimum speed of 2 mph when loaded uniformly with a load equal to 150 percent of truck capacity without any evidence of failure or damage to any part of the truck. This capability shall be determined by the test specified in 4.5.2.3.

3.6.4 End rack strength. The racks including sockets, shall be capable of withstanding without failure and permanent deformation, a minimum horizontal (with the floor surface) force of 300 pounds applied at the midpoint of the rack. This requirement shall be determined by the test specified in 4.5.2.4.

3.7 Lubrication fittings. Grease lube fittings shall conform to SAE J534 and shall be of the threaded types. The casters shall be grease packed with the manufacturer's recommended lubricant containing a rust inhibitor.

3.8 Repair and maintenance. Provisions shall be made in the design of the truck to provide replacement of hitch, hitch retainer racks, and caster wheels.

3.9 Identification marking. The truck shall be identified in accordance with MIL-STD-130, and shall include the manufacturer's name or trademark, part number, and the capacity of the truck. The marking shall be applied to the truck on plates conforming to MIL-P-514, Type I, Style 1, Composition C of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.10 Treatment and painting. Unless otherwise specified (see 6.2), the frame and deck of the truck shall be cleaned, treated and painted in accordance with MIL-T-704, Type A, except that the finish coat shall conform to TT-E-489 enamel, gloss yellow, Color Number 13538 of FED. STD. No. 595.

3.11 Workmanship. All parts, components, and assemblies of the truck including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs and sharp edges. No parts shall be damaged or impaired in any way and parts shall show no evidence of rust or corrosion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

3.11.1 Steel fabrication. Steel used in the fabrication of the truck shall provide original quality surface finish and shall be free from kinks and sharp bends. Steel having an eroded surface is not acceptable. The forming of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Burned surfaces of flame-cut material shall be free of slag. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly except where heat treatment is required. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.11.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the truck to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16-inch unless appropriate bridging techniques are used. The welding process used in fabrication of the truck shall be at the option of the contractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

4.3 Preproduction inspection.

4.3.1 Examination. The truck shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The truck shall be tested as specified in 4.5.2.1 through 4.5.2.5. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Level I.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.2. AQL shall be 2.5 percent defective for all tests.

4.5 Inspection procedure.

4.5.1 Examination. Samples shall be examined for defects of the characteristics that are listed in Table II.

Table II. Examination Schedule

Number	Characteristic	Requirement Paragraph
Major		
101.	Material not as specified.	3.3
102.	Capacity not as specified.	3.4
103.	Class 1 or Class 2 truck does not consist of parts as specified.	3.5.1, 3.5.2
104.	Frame and deck not as specified.	3.5.3
105.	Sockets not as specified.	3.5.4
106.	Caster support not as specified.	3.5.5
107.	End racks not as specified.	3.5.6
108.	Hitch and hitch retainers not as specified.	3.5.7
109.	Coupler not as specified.	3.5.8
110.	Running gear not as specified.	3.5.9
111.	Fastening devices not as specified.	3.5.11
112.	Lubrication fittings not as specified.	3.7

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

Table II. Examination Schedule (Cont'd)

Number	Characteristic	Requirement Paragraph
113.	Steel fabrication not as specified.	3.11.1
114.	Welding not as specified.	3.11.2
Minor		
201.	Repair and maintenance not as specified.	3.8
202.	Identification not as specified.	3.9
203.	Treatment and painting not as specified.	3.10
204.	Workmanship not as specified.	3.11

4.5.2 Tests.

4.5.2.1 Maximum static load. The truck shall be placed on a dry, smooth, level, concrete surface. A load equal to two times the rated capacity of the truck shall be uniformly distributed over the entire deck surface. The load shall remain on the deck for a period of 1 hour after which the load shall be removed and the truck shall be inspected to determine conformance to the requirement of 3.6.1. Nonconformance to the requirements of 3.6.1 shall constitute failure of this test.

4.5.2.2 Rolling resistance. The truck shall be placed on a dry, smooth, level concrete surface. The truck shall be loaded with a uniform load equal to the rated capacity of the truck and the load shall be distributed equally on the deck. The truck, with load, shall then be accelerated by means of a suitable towing device employing a (tension) dynamometer. After the truck has attained a minimum speed of 2 mph, the maximum gage reading of the dynamometer shall be determined in an interval of 25 feet of truck travel to determine conformance to the requirements of 3.6.2. Nonconformance to the requirements of 3.6.2 shall constitute failure of this test.

4.5.2.3 Dynamic load. The truck shall be placed on a 3-inch-high platform. A load equal to 150 percent of rated capacity of the truck shall be uniformly distributed on the entire platform. The loaded truck shall be pushed off the 3-inch-high step onto a 1-inch-high step onto the floor as follows: (1) 10 times perpendicular to the platform edge and (2) 10 times at a 45-degree angle to the platform edge. The lower and upper surface forming the steps shall be smooth, level, and hard (concrete or steel) and the edges shall be parallel. After completion of the test, remove the load

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

and examine the truck to determine conformance to the requirements of 3.6.3. Evidence of failure or damage and nonconformance to the requirements of 3.6.3 shall constitute failure of this test.

4.5.2.4 End rack strength. The truck, with end racks assembled to the sockets shall be placed on a hard, level surface. The wheels shall be blocked for this test. A horizontal force shall be applied at the center of the end rack through a (tension) dynamometer. The applied force shall be in accordance with 3.6.4 and shall be applied for a period of 5 minutes. After removal of the applied force, the end racks and sockets shall be inspected to determine conformance to the requirements of 3.6.4. Nonconformance to the requirements of 3.6.4 shall constitute failure of this test.

4.5.2.5 Load distribution. The empty truck shall be weighed on a level floor scale and the weight recorded. Next, the empty truck shall be rolled off the scale so that only one pair of casters rest on the scale. A reading on the scale shall then be taken and recorded. Next, rated load shall be distributed equally on the entire loading surface of the deck and the reading on the scale shall be taken and recorded. Calculations shall then be performed utilizing the above data (weights) to determine conformance to the load distribution on the casters specified in 3.5.9. Nonconcurrence to the requirements of 3.5.9 shall constitute failure of the test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, Level I.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for defects of the characteristics listed in Table III. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

Table III. Examination Schedule

Number	Characteristic	Requirement Paragraph
Major		
115.	Packing not as specified.	5.1
Minor		
205.	Military marking not in accordance with MIL-STD-129.	5.2.1
206.	Commercial marking not in accordance with MIL-STD-1188.	5.2.2

5. PACKAGING

5.1 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A. Trucks to be delivered in quantity to a single destination shall be arranged in the storage configuration as specified in 3.5.12, and then packed in a close-fitting box conforming to MIL-B-26195, Type II, style optional, Class 1 or 2, with rubbing strips and plywood superstructure. Trucks to be delivered individually shall be packed in the most compact manner in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional. Blocking, bracing, and anchoring of the contents within each box shall be adequate to prevent movement or damage. Box closure shall be in accordance with the applicable box specification.

5.1.2 Level B. Trucks shall be packed as specified in 5.1.1 except boxes shall be Type I or Domestic Type as applicable.

5.1.3 Commercial. Trucks shall be packed in accordance with MIL-STD-1188.

5.2 Marking.

5.2.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.2.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. Trucks described herein are intended for carrying light and medium duty items over short distances. Trucks are intended for use in commissaries for storage and handling of food items. Trucks are also intended for use in warehouses or other medium duty work areas.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class truck required (see 1.2).
(c) Time frame required for submission of preproduction model (see 3.2.1).
(d) Material other than specified (see 3.3).
(e) Treatment and painting other than specified (see 3.10).
(f) The level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production trucks from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish trucks conforming to this specification.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activities:
DLA - GS
Army - AT

Project 3920-A148

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-T-52952(ME) Truck, Hand, Platform, 3,000 Pound Capacity

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research and
Development Command
ATTN: DRDME-DS
Fort Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:41Z
Check the source to verify that this is the current version before use.